

WEB CISC3003

A Deep Dive into RWD & Rate Limiting Strategies

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| Stability and Usability: The Pillars

External: User-Centric RWD

Focusing on **Task Completion** across all devices (phones, tablets, and desktops) to ensure a seamless rental experience.

Internal: System Integrity

Implementing **Rate Limiting** to safeguard the server from crashes, accidental double-clicks, and malicious bot traffic.

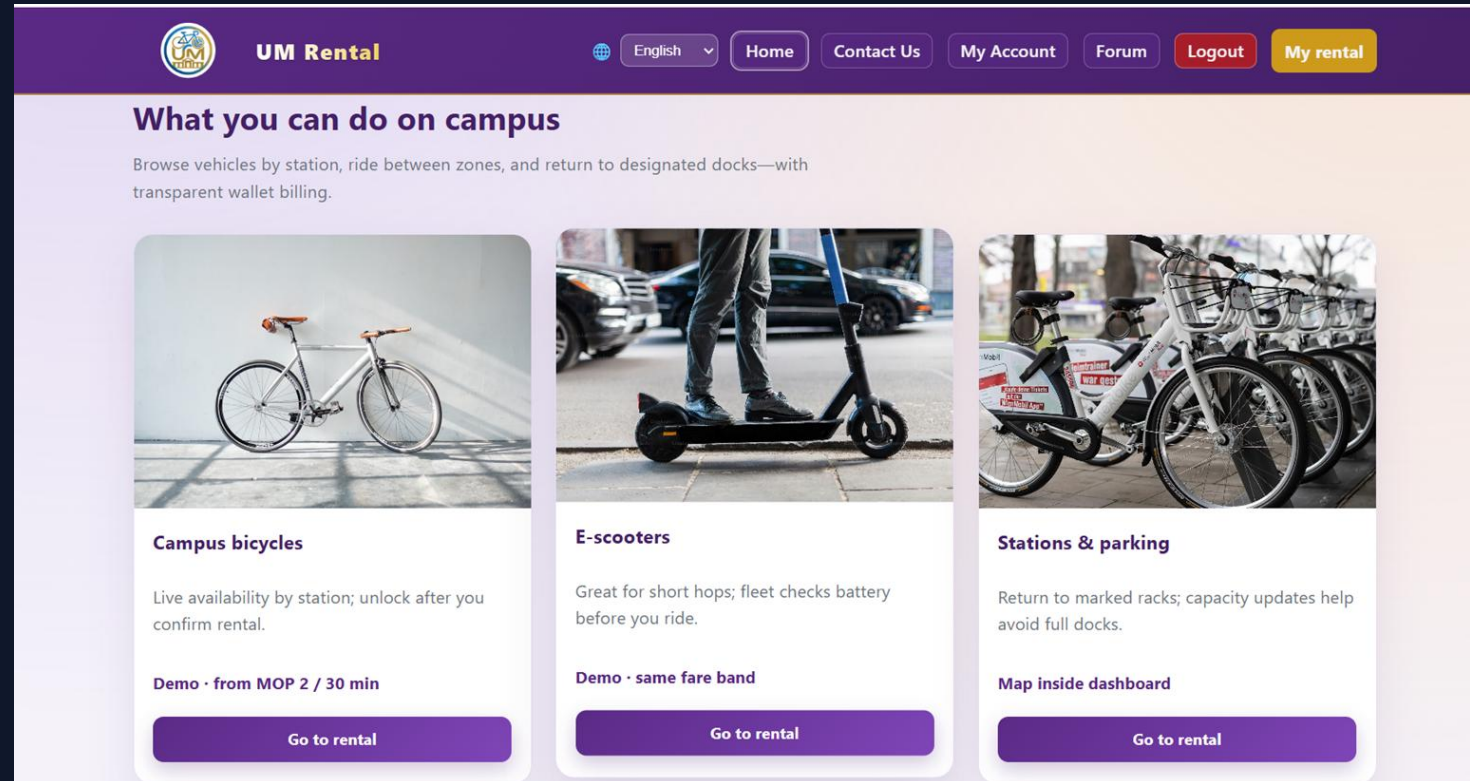
Pillar I: Responsive Web Design

Moving beyond "resizing" to functional excellence

Designing for Task Completion

We prioritized the core user journey to ensure 100% functionality on small screens:

- ➔ Secure & Clear Authentication
- 🚲 Real-time Vehicle Discovery
- 🕒 Seamless Rental Activation
- 📄 Intuitive Order Management



Strategic Layout Management



Mobile-First

Most important action buttons prioritized at the top for vertical thumb-scrolling efficiency.



Fluid Grids

Content reshuffles from side-by-side (Desktop) to top-to-bottom (Mobile) without losing context.



DRY Code

Shared style sheets ensure visual consistency and easy future scalability across the codebase.

Pillar II: Rate Limiting

The invisible gatekeeper of system performance

The Strategic Security Gate

Rate limiting serves as our first line of defense, checking request frequency before they ever hit the core application logic.



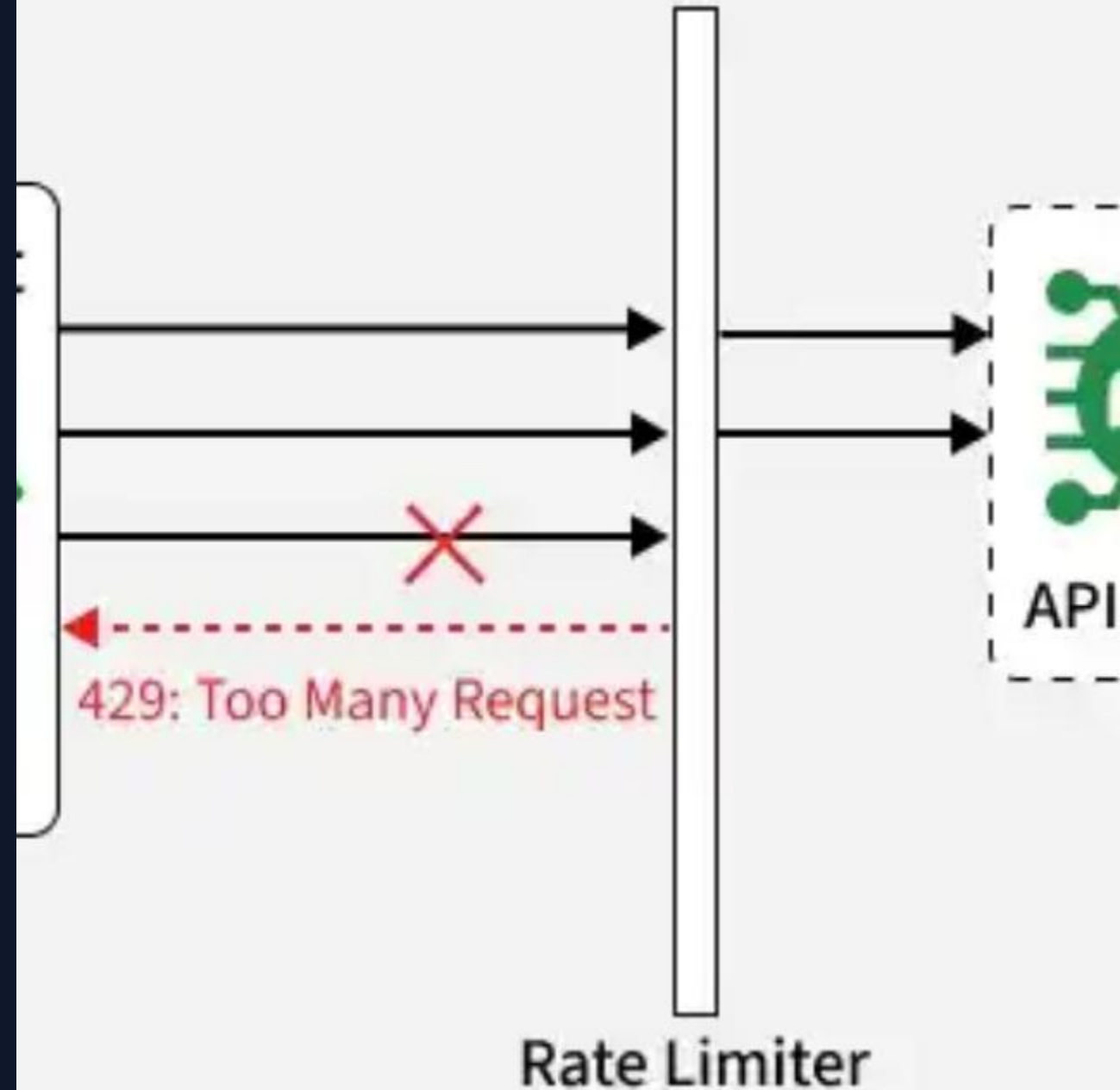
Early Detection: Identifies rapid-fire clicks instantly.



Throttling: "Wait a moment" messaging for safety.



Resource Guard: Keeps the server efficient.



Protecting Critical Infrastructure

Protected Action	Potential Risk	Mitigation Strategy
Login Requests	Brute Force / Credential Stuffing	Exponential back-off for failed attempts.
"Rent Now" Clicks	Accidental double-clicks / Race conditions	Request throttling per user session.
API Dashboard Load	Bot Scraping / Excessive Refreshing	Fixed-window rate limiting on API endpoints.

| Synergy: User & Server Harmony

99%

System Stability

Total Quality Management

RWD clarifies the interface, reducing user error and "panic clicks." Rate limiting catches the remaining edge cases.

Result: A "Super Strong" application that handles high traffic gracefully while remaining effortless for the end user.

Technical Project Repository

Category	Key Files / Evidence	Core Logic
RWD / CSS	style.css, dashboard.css, login.css	Media queries, Flexbox, Adaptive Layouts
Rate Limiting	config.php, api/login.php, rental_action.php	Throttling logic, Action protection gates
API Protection	api/dashboard_api.php, api/_common.php	Centralized request filtering chain

Image Sources



<https://cdn.dribbble.com/userupload/44859014/file/6fbfc3b2c2695646ca123b6b8c01aad9.png?crop=0x0-2395x1796&format=webp&resize=400x300&vertical=center>

Source: dribbble.com



https://media.geeksforgeeks.org/wp-content/uploads/20260331153020903963/what_is_rate_limiting_11.webp

Source: www.geeksforgeeks.org



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